

Down Payments, Deferred Homes: How LTV Restrictions Reshape Household Consumption

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Housing markets and consumption are tightly linked

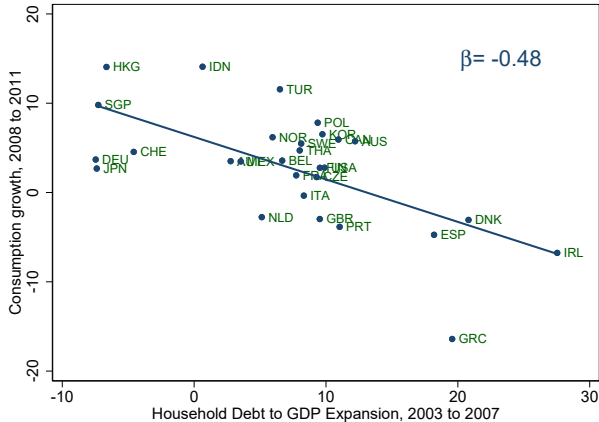
- House prices affect consumption via wealth and collateral channels
(Campbell & Cocco 2007; Mian, Rao & Sufi 2013; Guren et al. 2021)
- This link is one of the main drivers of business-cycle fluctuations and credit-driven booms
(Bernanke & Gertler 1986; Kiyotaki & Moore 1997; Iacoviello 2005)

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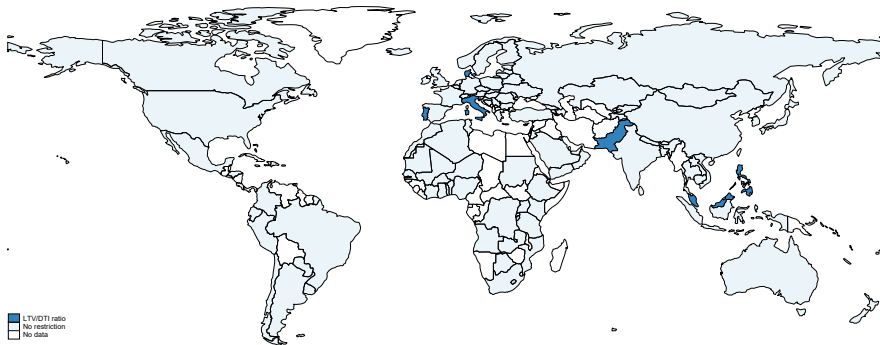
Via this link, household leverage amplifies the recessions

Housing and consumption



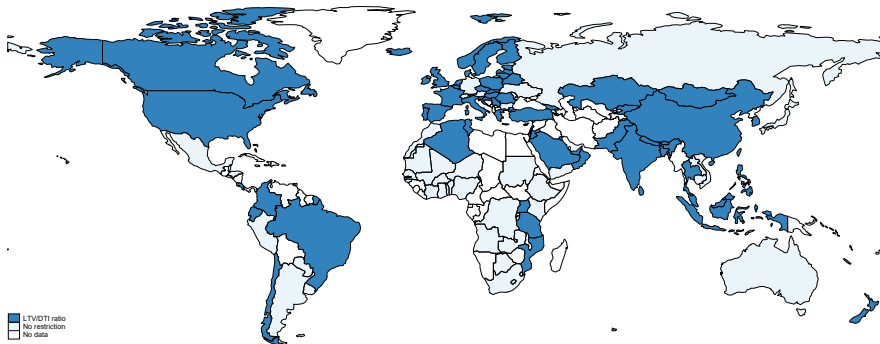
Increase in HH debt before the GFC predicts a decline in consumption during GFC

Macprudential Policies in 2000



Many countries have implemented borrowing restrictions on HH

Macprudential Policies in 2021



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Theoretical mechanism

How do prospective homebuyers (renters) adjust consumption in response to tighter borrowing constraints?

Important to understand the aggregate demand implications of these constraints

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Standard life-cycle models with binding borrowing constraints

- Tighter LTV $\Rightarrow$ higher down payment
- Renters *increase* savings to meet requirement
- Consumption *falls*

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Models with consumption smoothing and endogenous tenure choice

- Tighter LTV $\Rightarrow$ larger required savings
- Renters *delay* their purchase and saving because they expect higher income in the future
- Consumption *rises*

Balke et al. 2024

$\Rightarrow$ **Fundamentally different implications for aggregate demand**

- **Renters increase their consumption after the introduction of LTV restriction**
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 - Larger effects when earnings potential is high controlling for current income

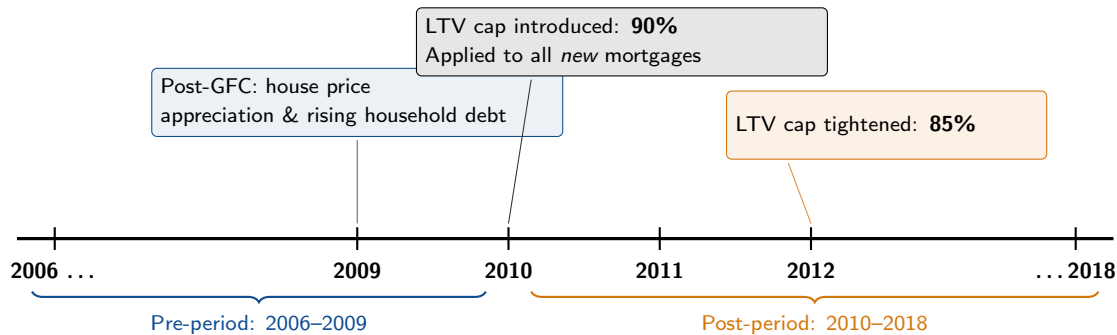
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- **Mechanism:** Delayed home purchase with delayed savings
 - Larger effects when earnings potential is high controlling for current income
- **Homebuyers** decrease their consumption before and after a home purchase
 - Evidence for rebuilding liquidity buffers

Data & Empirical Strategy

- Consumption, housing transactions, tax filings, individual characteristics
→ Consumption: Debit card transactions, digital invoice, direct remittances.
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 - Consumption: Debit card transactions, digital invoice, direct remittances. Approximately 80% of the card payments. 22 COICOP categories
- Sample: 2006-2018, Annual, Household-level
- **3 groups:** Renters, homebuyers, homeowners
 - Renters: Do not have housing wealth, no housing transactions before the restriction
 - Homebuyers: First-time homebuyers
 - Homeowners: Have housing wealth before 2006

The Norwegian LTV Restriction



- LTV regulation introduced in **2010**
- Restriction applies only to *new* mortgages $\Rightarrow$ existing homeowners unaffected

$$y_{it} = \beta Renter_i \times Post_t + \gamma_1 Renter_i + \gamma_2 Post_t + \epsilon_{it}$$

- y_{it} = Consumption / Income
- $Renter_i$: =1 if HH is a renter until 2010, 0 if HH is already a homeowner prior to 2006
→ Renters include HHs that purchase a home after the restriction
- $Post_t$: =1 if year $\geq$ 2010

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- $Renter_i$: =1 if HH is a renter until 2010, 0 if HH is already a homeowner prior to 2006
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- $Post_t$: =1 if year ≥ 2010
- **Identifying assumption**
 1. Without the restriction, the consumption/income difference between renters and homeowners would be the same
→ The levels can be different
 2. Homeowners are not affected by the restriction
→ The restriction can reduce homeowners' access to credit, or housing wealth

Results

Consumption Reaction of Renters

	Consumption/Income		
	(1)	(2)	(3)
Renter $\times$ Post	0.0318*** (0.00349)	0.0296*** (0.00374)	0.0276*** (0.00340)
Renter	-0.0837*** (0.00286)	-0.0814*** (0.00271)	
Post	-0.0168*** (0.00615)		
<i>Fixed Effects:</i>			
Year FE		✓	✓
Household FE			✓
Obs.	7,601,189	7,601,189	7,601,189
R ²	0.003	0.006	0.412
Mean(Dependent Var.)	0.662		

Renters increase their consumption after the LTV restriction

Do homeowners form a good control group for renters?

Possible issues

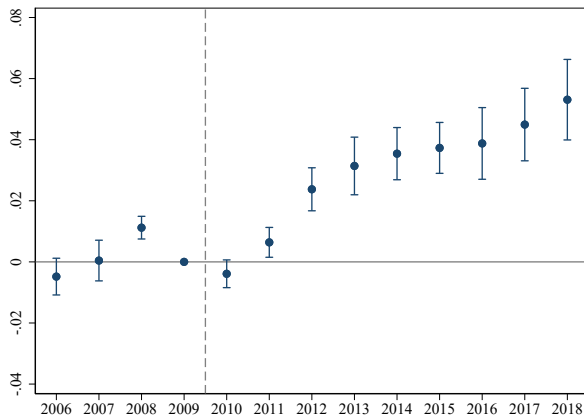
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→ **Parallel trends** before the restriction?

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	Consumption/Income					
	(1)	(2)	(3)	(4)	(5)	(6)
Renter $\times$ Post	0.0316*** (0.00451)	0.0293*** (0.00383)	0.0261*** (0.00374)	0.0412*** (0.00441)	0.0282*** (0.00361)	0.0385*** (0.00412)
<i>Fixed Effects:</i>						
Household FE	✓	✓	✓	✓	✓	✓
Age $\times$ Year FE	✓					
Deposit bins $\times$ Year FE		✓				
Financial Asset bins $\times$ Year FE			✓			
Total wealth bins $\times$ Year FE				✓		
Parental wealth bins $\times$ Year FE					✓	
Full Interacted FE						✓
Obs.	7,601,189	7,601,189	7,601,189	7,601,189	7,601,189	7,480,133
R ²	0.413	0.412	0.412	0.413	0.412	0.445
Mean(Dependent Var.)	0.660					

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	Consumption/Income				
	(1) LTV<95%	(2) LTV<85%	(3) LTV<75%	(4) LTV<65%	(5)
Renter $\times$ Post	0.0258*** (0.00314)	0.0252*** (0.00302)	0.0242*** (0.00289)	0.0230*** (0.00274)	0.0308*** (0.00309)
<i>Fixed Effects:</i>					
Household FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	
Municipality $\times$ Year FE					✓
Obs.	7,230,463	7,062,467	6,809,292	6,444,576	7,601,189
R ²	0.411	0.411	0.413	0.415	0.417
Mean(Dependent Var.)	0.656				

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1. Homeowners are different from renters
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3. The effect of homeownership on consumption changes over time
 - Housing wealth or house-related consumption
 - **Solution**: Within-renter comparison!

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 - **Solution**: Within-renter comparison!

Within-renter comparison

- Are there renters who are less exposed to the restriction?
 - Renters want to remain as renters or are very similar to existing homeowners
 - How do we detect such renters with high ownership capacity?
 1. Split the sample into two: homeowners and renters
 2. Use XGBoost to predict the probability of being a homeowner by using 2006 information
- **Intuition**: **Renters with a high probability** have observables suggesting that they are similar to homeowners but still do not buy. Hence, they **are less affected**.

Within Renter Comparison

$$y_{it} = \beta (1 - \text{Homeowner Prob})_i \times \text{Post}_t + \gamma_1 (1 - \text{Homeowner Prob})_i + \gamma_2 \text{Post}_t + \epsilon_{it}$$

	Consumption/Income			
	(1)	(2)	(3)	(4)
(1-Homeowner prob.) $\times$ Post	0.0489*** (0.00650)	0.0320*** (0.00743)		
1(Homeowner prob.<80 _{pct}) $\times$ Post			0.0299*** (0.00523)	0.0173*** (0.00523)
<u>Fixed Effects:</u>				
Year FE	✓	✓	✓	✓
Household FE	✓	✓	✓	✓
Matching		✓		✓
Obs.	433,427	146,809	484,520	155,020
R ²	0.444	0.440	0.448	0.441
Mean(Dependent Var.)	0.606			

Results hold when making use of heterogeneity within renters

Why do renters increase their consumption?

- The aim of the policy is to decrease household leverage by increasing saving rates
→ Yet, we find the opposite effect! Why?

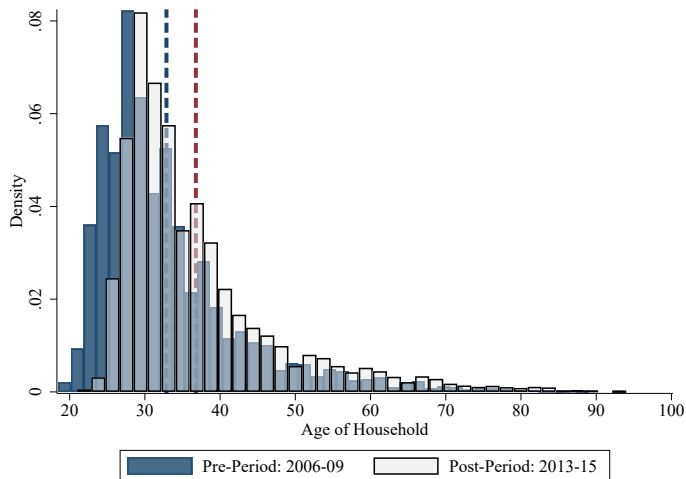
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- **Renters may delay their home purchase and savings** (Karlman et al. (2024))
 - HHs need to reduce their consumption to accumulate savings
 - Leads to deviation in consumption smoothing, which is costly for renters
 - The disutility of this deviation can be larger than the benefits of an earlier home purchase
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- **Two predictions**
 1. There must be a delay in home purchases
 2. The increase in consumption must be stronger for renters who expect a larger increase in their earnings

Age Distribution of Home Buying Cohorts



Homebuying cohorts get older

Earnings Potential Gap

Idea: How far a household is from the upper end of the income distribution among comparable households who are later in their career stages.

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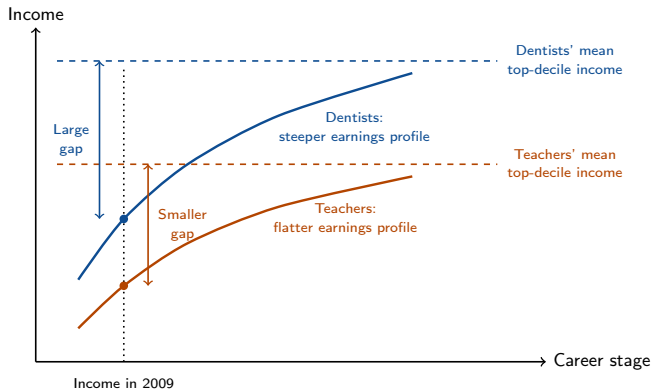
Comparable households:
education $\times$ labor-market
region $\times$ immigration background

$$\text{Gap}_i = \text{Top-decile income}_c - \text{Income}_{i,2009}$$

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Comparable households:
education × labor-market
region × immigration background



$$\text{Gap}_i = \text{Top-decile income}_c - \text{Income}_{i,2009}$$

The gap is higher for HHs who are in the early stages of their careers and HHs whose cells have higher frontier income

Earnings Potential Gap

	Above top 10%	1 st Quart.	2 nd Quart.	3 rd Quart.	4 th Quart.
	(1)	(2)	(3)	(4)	(5)
Renter $\times$ Post	-0.00160 (0.00822)	0.00860*** (0.00297)	0.0169*** (0.00441)	0.0317*** (0.00494)	0.0401*** (0.00356)
<i>Fixed Effects:</i>					
Household FE	✓	✓	✓	✓	✓
Cell $\times$ Year FE	✓	✓	✓	✓	✓
Income bins $\times$ Year FE	✓	✓	✓	✓	✓
Obs.	544,254	1,690,597	1,866,771	1,822,874	1,668,886
R ²	0.424	0.407	0.434	0.465	0.470
Mean(Dependent Var.)	0.662				

Renters with greater anticipated income growth are more likely to delay the costly deviation from consumption smoothing

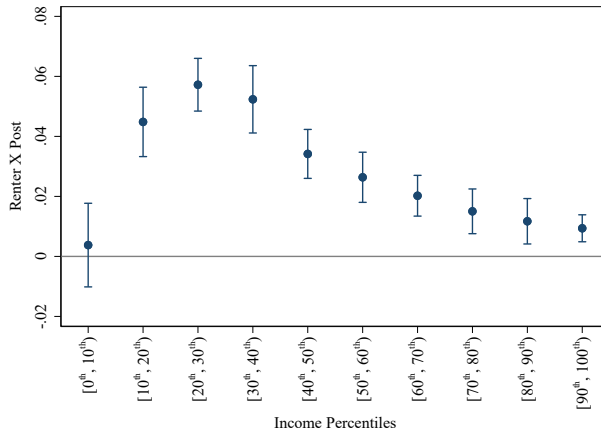
- The effect is stronger in regions with higher LTV ratios measured before the restriction
- The effect is insignificant for the lowest income decile. But stronger for the rest of low-income renters
- The renters increase their luxury spending, measured by expenditure elasticity
- No evidence for giving-up, as renters increase their effort in the labor market
- Renters move less and live in neighborhoods with higher income levels
- Renters increase their debt and reduce savings (deposits and financial assets)

- What about renters that purchased a home under the LTV restriction?
- We compare homebuyers in 2013 to homebuyers in 2009 and use homeowners as additional control for time effects
- Homebuyers in 2013 reduce their consumption significantly before the home purchase
- They also keep their consumption low after the purchase
 - They accumulate savings faster
 - No evidence for purchasing another home or locational sorting

- **Financial stability implications of macroprudential restrictions are not specific to homeowners**
- **LTV restrictions may stimulate short-term demand by increasing renters' consumption**

Thank You!

Income Heterogeneity - Levels



The effect is stronger for households with lower income

	MuniLTV Interaction	MuniLTV <25th perc.	MuniLTV 25th–75th perc.	MuniLTV >75th perc.
	(1)	(2)	(3)	(4)
Renter × Post × Municipality LTV	0.0036** (0.00177)			
Renter × Post	0.0306*** (0.00257)	0.0206*** (0.00422)	0.0289*** (0.00399)	0.0377*** (0.00314)
Post × Municipality LTV	0.0134*** (0.00214)			
<i>Fixed Effects:</i>				
HH FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Obs.	7,058,502	1,701,527	3,496,391	2,403,271
R ²	0.413	0.415	0.412	0.410
Mean(Municipality LTV)	0.923	0.810	0.921	1.030
Diff. to MuniLTV > 75th Coeff		0.0171***	0.0087*	
Diff. to MuniLTV < 25th Coeff			0.0084	

Renters that live in municipalities with higher LTV respond more strongly

Consumption Categories

	Cons. type/Income		Cons. type/Total Cons.	
	(1)	(2)	(3)	(4)
	$\epsilon < 1$	$\epsilon > 1$	$\epsilon < 1$	$\epsilon > 1$
Renter $\times$ Post	0.00324 (0.00252)	0.0199*** (0.00122)	-0.0192*** (0.00310)	0.0211*** (0.00363)
<i>Fixed Effects:</i>				
Household FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Obs.	7,601,189	7,601,189	7,601,189	7,601,189
R ²	0.579	0.444	0.547	0.542
Mean(Dependent Var.)	0.147	0.380	0.301	0.687

Substitution towards elastic consumption categories